T3 E_-2: full general UV Lagrangian and RGEs

Matchete/RGBeta symbolic output in mathematical notation

E_-2

Additional quartics: 6, additional bilinears: 1, extra Yukawa: y2Conj.
General matching and Wilson coefficients are not computed in this symbolic report.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Complete BSM UV Lagrangian

Matchete BSM Lagrangian, excluding the pure Standard Model contribution. $\mathcal{L}_{\text{BSM}} = \text{Coupling(lambdaT3ExtraM000002I1, \{\}, 0)CG(T3ExtraCGM000002I1, \{Index(d\$1, T3BSMd3), Inde}$

$\text{+Coupling(lambdaT3ExtraQ000004I1, \{\}, 0)Bar(Field(NewScalar2, Scalar, \{Index(d881, T3BSMd3)\}, \{\}))Bar(Field(NewScalar2, Scalar, \{Index(d882, T3BSMd3)\}, \{\}))Bar(Field(NewScalar2, Scalar, \{Index(d883, T3BSMd3)\}, \{\}))Bar(Field(NewScalar2, Scalar, \{Index(d884, T3BSMd3)\}, \{\}))CG(T3ExtraCGQ000004I1, \{Index(d881, T3BSMd3), Index(d882, T3BSMd3), Index(d883, T3BSMd3), Index(d884, T3BSMd3)\})}$

+Coupling(lambdaT3ExtraQ020101I1, {}, 0)Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])Bar(Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}])CG(T3ExtraCGQ020101I1, {Index(d$$1, SU2L(fund)), Index(d$$2, SU2L(fund)), Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})

+Coupling(lambdaT3ExtraQ110002I1, {}, 0)Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])CG(T3ExtraCGQ110002I1, {Bar(Index(d$$2, SU2L(fund))), Index(d$$1, SU2L(fund)), Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})

$$+\text{Coupling}(\text{MS1}, \{\}, 0)^2\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})) + \text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

+$\frac{1}{2}$Coupling(lambdaH1Inv1, {}, 0)Field(NewScalar1, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])CG(T3HiggsPortal1CGInv1, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3))

+$\frac{1}{2}$Coupling(lambdaH1Inv1, {}, 0)Field(NewScalar1, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])CG(T3HiggsPortal1CGInv1, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$2, T3BSMd3), Index(d$$1, T3BSMd3))

+$\frac{1}{2}$Coupling(lambdaH1Inv2, {}, 0)Field(NewScalar1, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])CG(T3HiggsPortal1CGInv2, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3))

+$\frac{1}{2}$Coupling(lambdaH1Inv2, {}, 0)Field(NewScalar1, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])CG(T3HiggsPortal1CGInv2, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$2, T3BSMd3), Index(d$$1, T3BSMd3))

+Coupling(lambdaT3, {}, 0)Field(NewScalar1, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}]Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]CG(T3MixCG, {Bar(Index(d$$1, SU2L(fund))), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$2, T3BSMd3), Index(d$$1, T3BSMd3))

+Coupling(lambdaT3ExtraQ001102I1, {}, 0)Field(NewScalar1, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {}))CG(T3ExtraCGQ001102I1, {Index(d$$4, T3BSMd3), Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3)})

+Coupling(lambdaT3ExtraQ001102I2, {}, 0)Field(NewScalar1, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {}))CG(T3ExtraCGQ001102I2, {Index(d$$4, T3BSMd3), Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3)})

+$\frac{1}{2}$Coupling(lambdaS1Inv1, {}, 0)Field(NewScalar1, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar1, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar1, Scalar, {Index(d$$2, T3BSMd3)}, {}))CG(T3Scalar1SelfCGInv1, {Index(d$$1, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$4, T3BSMd3)})

+$\frac{1}{2}$Coupling(lambdaS1Inv2, {}, 0)Field(NewScalar1, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar1, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar1, Scalar, {Index(d$$2, T3BSMd3)}, {}))CG(T3Scalar1SelfCGInv2, {Index(d$$1, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$4, T3BSMd3)})

$$+\text{Coupling}(\text{MS2}, \{\}, 0)^2\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})) + \text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

+Bar(Coupling(lambdaT3, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])Bar(Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}])CG(Bar(T3MixCG), {Index(d$$1, SU2L(fund)), Index(d$$2, SU2L(fund))}, Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3))

+$\frac{1}{2}$Coupling(lambdaH2Inv1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])CG(T3HiggsPortal2CGInv1, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3))

+$\frac{1}{2}$Coupling(lambdaH2Inv1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])CG(T3HiggsPortal2CGInv1, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$2, T3BSMd3), Index(d$$1, T3BSMd3))

+$\frac{1}{2}$Coupling(lambdaH2Inv2, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])CG(T3HiggsPortal2CGInv2, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3))

+$\frac{1}{2}$Coupling(lambdaH2Inv2, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])CG(T3HiggsPortal2CGInv2, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$2, T3BSMd3), Index(d$$1, T3BSMd3))

+Bar(Coupling(lambdaT3ExtraQ020101I1, {}, 0)Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}]Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]CG(Bar(T3ExtraCGQ020101I1), {Bar(Index(d$$1, SU2L(fund))), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3))

+Bar(Coupling(lambdaT3ExtraM000002I1, {}, 0))CG(T3ExtraCGM000002I1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})

+Bar(Coupling(lambdaT3ExtraQ110002I1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Field[H, Scalar, {Index(d$$2, SU2L(fund))}, {}]Bar(Field[H, Scalar, {Index(d$$1, SU2L(fund))}, {}])CG(T3ExtraCGQ110002I1, {Bar(Index(d$$2, SU2L(fund))), Index(d$$1, SU2L(fund)), Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})

+Coupling(lambdaT3ExtraQ000013I1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {}))CG(T3ExtraCGQ000013I1, {Index(d$$4, T3BSMd3), Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3)})

+Coupling(lambda12Inv1, {}, 0)Field(NewScalar1, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))CG(T3CrossScalarCGInv1, {Index(d$$1, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$4, T3BSMd3)})

+Coupling(lambda12Inv2, {}, 0)Field(NewScalar1, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))CG(T3CrossScalarCGInv2, {Index(d$$1, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$4, T3BSMd3)})

+Coupling(lambda12Inv3, {}, 0)Field(NewScalar1, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))CG(T3CrossScalarCGInv3, {Index(d$$1, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$4, T3BSMd3)})

+$\frac{1}{2}$Coupling(lambdaS2Inv1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))CG(T3Scalar2SelfCGInv1, {Index(d$$1, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$4, T3BSMd3)})

+$\frac{1}{2}$Coupling(lambdaS2Inv2, {}, 0)Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))CG(T3Scalar2SelfCGInv2, {Index(d$$1, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$4, T3BSMd3)})

+Bar(Coupling(lambdaT3ExtraQ001102I1, {}, 0)Field(NewScalar1, Scalar, {Index(d$$2, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))CG(T3ExtraCGQ001102I1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$4, T3BSMd3)})

+Bar(Coupling(lambdaT3ExtraQ001102I2, {}, 0)Field(NewScalar1, Scalar, {Index(d$$2, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))CG(T3ExtraCGQ001102I2, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$4, T3BSMd3)})

+Bar(Coupling(lambdaT3ExtraQ000013I1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))CG(T3ExtraCGQ000013I1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$4, T3BSMd3)})

+Bar(Coupling(lambdaT3ExtraQ000004I1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})CG(T3ExtraCGQ000004I1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$4, T3BSMd3)})

$$-\text{Coupling}(\text{MF}, \{\}, 0)\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$$

+Coupling(y2, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))CG(T3Y2CG, {Index(d$$1, SU2L(fund)), Bar(Index(d$$1, T3BSMd2))}, Index(d$$1, T3BSMd3))NCM(Bar(Field[I, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {}]), DiracProduct(Proj(1)), Field[NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}])

+Coupling(y2Conj, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})CG(T3Y2ConjCG, {Index(d$$1, SU2L(fund)), Bar(Index(d$$1, T3BSMd2))}, Index(d$$1, T3BSMd3))NCM(Bar(Field[I, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {}]), DiracProduct(Proj(1)), Field[NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}])

+Coupling(y1, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {})CG(T3Y1CG, {Index(d$$1, SU2L(fund)), Index(d$$1, T3BSMd2), Index(d$$1, T3BSMd3)})NCM(Bar(Field[I, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {}]), DiracProduct(GammaCC, Proj(1)), Transp(Bar(Field[NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}])))

$$+i\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{GammaM}(\text{Index}(\text{d\$\$1}, \text{Lorentz}))), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

+Bar(Coupling(y2Conj, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))CG(Bar(T3Y2ConjCG), {Bar(Index(d$$1, SU2L(fund))), Index(d$$1, T3BSMd2), Index(d$$1, T3BSMd3)})NCM(Bar(Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {})), DiracProduct(Proj(-1)), Field[I, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {}])

+Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Bar(Coupling(y2, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)CG(Bar(T3Y2CG), {Bar(Index(d$$1, SU2L(fund))), Index(d$$1, T3BSMd2), Index(d$$1, T3BSMd3)})NCM(Bar(Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {})), DiracProduct(Proj(-1)), Field[I, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {}])

+Bar(Coupling(y1, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))CG(Bar(T3Y1CG), {Bar(Index(d$$1, SU2L(fund))), Bar(Index(d$$1, T3BSMd2)), Index(d$$1, T3BSMd3)})NCM(Transp(Field[I, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {}]), DiracProduct(GammaCC, Proj(-1)), Field[NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}])

Interaction inventory

- `Yukawa1_GenericCG`
- `Yukawa2_GenericCG`
- `Yukawa2ConjugateScalar_GenericCG`
- `Scalar1SelfInv1`

- Scalar1SelfInv2
- Scalar2SelfInv1
- Scalar2SelfInv2
- HiggsPortal1Inv1
- HiggsPortal1Inv2
- HiggsPortal2Inv1
- HiggsPortal2Inv2
- ScalarCrossPortalInv1
- ScalarCrossPortalInv2
- ScalarCrossPortalInv3
- T3ScalarMix_GenericCG
- T3ExtraQ_0_0_0_0_0_4_1
- T3ExtraQ_0_0_0_0_1_3_1
- T3ExtraQ_0_0_1_1_0_2_1
- T3ExtraQ_0_0_1_1_0_2_2
- T3ExtraQ_0_2_0_1_0_1_1
- T3ExtraQ_1_1_0_0_0_2_1
- T3ExtraM_0_0_0_0_0_2_1

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{59{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = \frac{{g_2}^3}{6}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = & \, 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = & \, \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$
\begin{aligned}
16\pi^2 \frac{dY_e}{d\ln\mu} = & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\
& + \frac{3}{4}(\Big(y_1\ (y_1)^\dagger\ Y_e\Big)_{ij} + \Big(y_2\ (y_2)^\dagger\ Y_e\Big)_{ij} + \Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\ Y_e\Big)_{ij} + 2\,\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\
& - \frac{15}{4}{g_1}^2\,(Y_e)_{ij}
\end{aligned}
$$

y_1

$$
\begin{aligned}
16\pi^2 \frac{dy_1}{d\ln\mu} = & T_\nu^{(1)}\,(y_1)_{ij} \\
& + \frac{1}{4}(3\,\Big(y_2\ (y_2)^\dagger\ y_1\Big)_{ij} + 3\,\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\ y_1\Big)_{ij} + 2\,\Big(Y_e\ (Y_e)^\dagger\ y_1\Big)_{ij} + 6\,\Big(y_1\ (y_1)^\dagger\ y_1\Big)_{ij}) \\
& - \frac{9}{2}{g_2}^2\,(y_1)_{ij} \\
& - \frac{3}{2}{g_1}^2\,(y_1)_{ij}
\end{aligned}
$$

y_2

$$
\begin{aligned}
16\pi^2 \frac{dy_2}{d\ln\mu} = & \mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)\,(y_2)_{ij} \\
& + 2\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big)\,(\mathrm{y2Conj})_{ij} \\
& + T_\nu^{(2)}\,(y_2)_{ij} \\
& + \frac{1}{12}(9\,\Big(y_1\ (y_1)^\dagger\ y_2\Big)_{ij} + 29\,\Big(y_2\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\Big)_{ij} + 29\,\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\ y_2\Big)_{ij} + 6\,\Big(Y_e\ (Y_e)^\dagger\ y_2\Big)_{ij} + 18\,\Big(\\
& - \frac{9}{2}{g_2}^2\,(y_2)_{ij} \\
& - \frac{3}{2}{g_1}^2\,(y_2)_{ij}
\end{aligned}
$$

M_F

$$
\begin{aligned}
16\pi^2 \frac{dM_F}{d\ln\mu} = & \frac{3}{4}(\Big(\left(y_1\right)^T\ (y_1)^*\ M_F\Big)_{ij} + \Big(M_F\ (y_2)^\dagger\ y_2\Big)_{ij} + \Big(M_F\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\Big)_{ij}) \\
& - \frac{9}{2}{g_2}^2\,(M_F)_{ij} \\
& - \frac{3}{2}{g_1}^2\,(M_F)_{ij}
\end{aligned}
$$

m_1^2

$$
16\pi^2 \frac{dm_1^2}{d\ln\mu} = m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})
$$

$$
\begin{aligned}
&- 4\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)\\
&+ 4\mathrm{muT3GeneralS2Sq}(3\,(\mathrm{lambdaT3GeneralQ001102I1})^* + (\mathrm{lambdaT3GeneralQ001102I2})^*)\\
&+ 12\mathrm{lambdaT3GeneralQ001102I1}\,(\mathrm{muT3GeneralS2Sq})^*\\
&+ 4\mathrm{lambdaT3GeneralQ001102I2}\,(\mathrm{muT3GeneralS2Sq})^*\\
&- 12{g_2}^2 m_1^2\\
&- 6{g_1}^2 m_1^2\\
&+ m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})
\end{aligned}
$$

m_2^2

$$
\begin{aligned}
16\pi^2\,\frac{dm_2^2}{d\ln\mu} = {}& m_2^2(2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})\\
&+ \mathrm{muT3GeneralS2Sq}(4\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big) + 20\,(\mathrm{lambdaT3GeneralQ000013I1})^*)\\
&- 4\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)\\
&- 4\,\mathrm{Tr}\Big(M_F\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\ (M_F)^\dagger\Big)\\
&+ 4\,(\mathrm{muT3GeneralS2Sq})^*\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big)\\
&+ 20\mathrm{lambdaT3GeneralQ000013I1}\,(\mathrm{muT3GeneralS2Sq})^*\\
&- 12{g_2}^2 m_2^2\\
&+ m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})
\end{aligned}
$$

λ_1

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_1}{d\ln\mu} = {}& \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))\\
&- 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)\\
&- 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)\\
&+ 4\lambda_5\,(\lambda_5)^*\\
&+ 4\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^*\\
&+ 24\mathrm{lambdaT3GeneralQ110002I1}\,(\mathrm{lambdaT3GeneralQ110002I1})^*\\
&+ \frac{{g_2}^4}{4}\\
&+ {g_1}^2\left(\frac{{g_2}^2}{2} - 3\lambda_1\right)\\
&- 9{g_2}^2\lambda_1\\
&+ \frac{{g_1}^4}{4}\\
&+ 12{\lambda_1}^2\\
&+ 6{\lambda_{HS_1}^{(1)}}^2\\
&+ 3{\lambda_{HS_1}^{(A)}}^2\\
&+ 6{\lambda_{HS_2}^{(1)}}^2\\
&+ 3{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = {} & \lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)}) \\
& - \frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big) \\
& + 8\mathrm{lambdaT3GeneralQ001102I1}(3\,(\mathrm{lambdaT3GeneralQ001102I1})^* + (\mathrm{lambdaT3GeneralQ001102I2})^*) \\
& + \mathrm{lambdaT3GeneralQ001102I2}\left(8\,(\mathrm{lambdaT3GeneralQ001102I1})^* + \frac{20\,(\mathrm{lambdaT3GeneralQ001102I2})^*}{3}\right) \\
& + 22{g_2}^4 \\
& + {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_1}^{(1)}\right) \\
& - 24{g_2}^2\lambda_{S_1}^{(1)} \\
& + 12{g_1}^4 \\
& + 6{\lambda_{12}^{(1)}}^2 \\
& + 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
& + \frac{20{\lambda_{12}^{(\times)}}^2}{3} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 14{\lambda_{S_1}^{(1)}}^2 \\
& + 16{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = {} & \lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) + 8\lambda_{S_2}^{(A)}) \\
& + \frac{32}{3}\,(\mathrm{lambdaT3GeneralQ000013I1})^*\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big) \\
& + \mathrm{lambdaT3GeneralQ000013I1}\left(\frac{32}{3}\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big) + \frac{352\,(\mathrm{lambdaT3GeneralQ000013I1})^*}{3}\right) \\
& - \frac{248}{45}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) \\
& - \frac{248}{45}\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\ (y_2)^\dagger\Big) \\
& - \frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& - \frac{8}{3}\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) \\
& + \frac{832}{3}\mathrm{lambdaT3GeneralQ000004I1}\,(\mathrm{lambdaT3GeneralQ000004I1})^* \\
& + \frac{8}{3}\mathrm{lambdaT3GeneralQ001102I1}(3\,(\mathrm{lambdaT3GeneralQ001102I1})^* + (\mathrm{lambdaT3GeneralQ001102I2})^*) \\
& + \mathrm{lambdaT3GeneralQ001102I2}\left(\frac{8\,(\mathrm{lambdaT3GeneralQ001102I1})^*}{3} + 4\,(\mathrm{lambdaT3GeneralQ001102I2})^*\right) \\
& + \frac{16}{3}\mathrm{lambdaT3GeneralQ110002I1}\,(\mathrm{lambdaT3GeneralQ110002I1})^* \\
& + 22{g_2}^4 \\
& - 24{g_2}^2\lambda_{S_2}^{(1)} \\
& + 6{\lambda_{12}^{(1)}}^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20{\lambda_{12}^{(\times)}}^2}{3} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 14{\lambda_{S_2}^{(1)}}^2 \\
&+ 16{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} &= \lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_1}^{(A)}) \\
&\quad - 2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
&\quad + 4\lambda_5\,(\lambda_5)^* \\
&\quad + 4\mathrm{lambdaT3GeneralQ110002I1}(3\,(\mathrm{lambdaT3GeneralQ001102I1})^* + (\mathrm{lambdaT3GeneralQ001102I2})^*) \\
&\quad + 12\mathrm{lambdaT3GeneralQ001102I1}\,(\mathrm{lambdaT3GeneralQ110002I1})^* \\
&\quad + 4\mathrm{lambdaT3GeneralQ001102I2}\,(\mathrm{lambdaT3GeneralQ110002I1})^* \\
&\quad + 4\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^* \\
&\quad + 6{g_2}^4 \\
&\quad - \frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
&\quad + 3{g_1}^4 \\
&\quad - \frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
&\quad + \lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
&\quad + 6\lambda_1\lambda_{HS_1}^{(1)} \\
&\quad + 4{\lambda_{HS_1}^{(1)}}^2 \\
&\quad + 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&\quad + 6{\lambda_{HS_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} &= \lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_2}^{(A)}) \\
&\quad + \mathrm{lambdaT3GeneralQ110002I1}(4\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big) + 4(5\,(\mathrm{lambdaT3GeneralQ000013I1})^* + 4\,(\mathrm{lambdaT3Gene} \\
&\quad + 4\,(\mathrm{lambdaT3GeneralQ110002I1})^*\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big) \\
&\quad - 2\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
&\quad - 2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
&\quad + 4\lambda_5\,(\lambda_5)^* \\
&\quad + 20\mathrm{lambdaT3GeneralQ000013I1}\,(\mathrm{lambdaT3GeneralQ110002I1})^* \\
&\quad + 4\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^* \\
&\quad + 6{g_2}^4
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2} \\
&- \frac{3{g_1}^2\lambda_{HS_2}^{(1)}}{2} \\
&+ \lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ 6{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} &= \lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}) \\
&+ \mathrm{lambdaT3GeneralQ001102I1}(4\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big) + 4(5\,(\mathrm{lambdaT3GeneralQ000013I1})^* + 4\,(\mathrm{lambdaT3Gener} \\
&+ 4\,(\mathrm{lambdaT3GeneralQ001102I1})^*\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big) \\
&- \frac{22}{15}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
&- \frac{22}{15}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) \\
&+ 2\lambda_5\,(\lambda_5)^* \\
&+ 4\mathrm{lambdaT3GeneralQ000013I1}(5\,(\mathrm{lambdaT3GeneralQ001102I1})^* + (\mathrm{lambdaT3GeneralQ001102I2})^*) \\
&+ 4\mathrm{lambdaT3GeneralQ001102I2}((\mathrm{lambdaT3GeneralQ000013I1})^* + (\mathrm{lambdaT3GeneralQ001102I2})^*) \\
&+ 2\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^* \\
&+ \frac{63{g_2}^4}{5} \\
&- 24{g_2}^2\lambda_{12}^{(1)} \\
&- 6{g_1}^2\lambda_{12}^{(1)} \\
&+ 4{\lambda_{12}^{(1)}}^2 \\
&+ \lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
&+ \lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
&+ 16{\lambda_{12}^{(A)}}^2 \\
&+ 4{\lambda_{12}^{(\times)}}^2 \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} &= \lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times} \right. \\
&+ 2\,(\mathrm{lambdaT3GeneralQ020101I1})^*\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big) \\
&+ 4\mathrm{lambdaT3GeneralQ001102I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^*
\end{aligned}
$$

$$
\begin{aligned}
&- 2\text{lambdaT3GeneralQ001102I2}\,(\text{lambdaT3GeneralQ020101I1})^*\\
&+ 8\text{lambdaT3GeneralQ110002I1}\,(\text{lambdaT3GeneralQ020101I1})^*\\
&- \frac{33{g_2}^2\lambda_5}{2}\\
&- \frac{9{g_1}^2\lambda_5}{2}\\
&+ 2\lambda_{12}^{(1)}\lambda_5\\
&+ 2\lambda_1\lambda_5\\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5\\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_1}^{(A)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = {}& \lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_1}^{(A)})\\
&- \frac{4}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- \frac{4}{3}\lambda_5\,(\lambda_5)^*\\
&- \frac{4}{3}\text{lambdaT3GeneralQ020101I1}\,(\text{lambdaT3GeneralQ020101I1})^*\\
&+ {g_1}^2\left(-4{g_2}^2 - \frac{15\lambda_{HS_1}^{(A)}}{2}\right)\\
&- \frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}\\
&+ 12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}\\
&+ 2\lambda_1\lambda_{HS_1}^{(A)}\\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}\\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = {}& \lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\text{y2Conj}\ (\text{y2Conj})^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_2}^{(A)})\\
&+ \frac{4}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- \frac{4}{3}\,\mathrm{Tr}\Big(\text{y2Conj}\ (\text{y2Conj})^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- \frac{8}{9}\lambda_5\,(\lambda_5)^*\\
&- \frac{4}{3}\text{lambdaT3GeneralQ020101I1}\,(\text{lambdaT3GeneralQ020101I1})^*\\
&- \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}\\
&- \frac{3{g_1}^2\lambda_{HS_2}^{(A)}}{2}\\
&+ 12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}\\
&+ 2\lambda_1\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{S_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(A)}}{d\ln\mu} = &\, 4\lambda_{S_1}^{(A)} T_\nu^{(1)} \\
&- \frac{2}{3}\operatorname{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big) \\
&- \frac{4}{3}\text{lambdaT3GeneralQ001102I2}\,(\text{lambdaT3GeneralQ001102I2})^* \\
&- 2{g_2}^4 \\
&+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_1}^{(A)}\right) \\
&- 24{g_2}^2\lambda_{S_1}^{(A)} \\
&+ 12{\lambda_{12}^{(A)}}^2 \\
&- \frac{4{\lambda_{12}^{(\times)}}^2}{3} \\
&+ 3{\lambda_{HS_1}^{(A)}}^2 \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 20{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(A)}}{d\ln\mu} = &\, \lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\operatorname{Tr}\Big(\text{y2Conj}\ (\text{y2Conj})^\dagger\Big)) \\
&- \frac{4}{3}\,(\text{lambdaT3GeneralQ000013I1})^*\operatorname{Tr}\Big(y_2\ (\text{y2Conj})^\dagger\Big) \\
&+ \text{lambdaT3GeneralQ000013I1}\left(-\frac{4}{3}\operatorname{Tr}\Big(\text{y2Conj}\ (y_2)^\dagger\Big) - \frac{44\,(\text{lambdaT3GeneralQ000013I1})^*}{3}\right) \\
&+ \frac{76}{45}\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ \text{y2Conj}\ (\text{y2Conj})^\dagger\Big) \\
&+ \frac{76}{45}\operatorname{Tr}\Big(y_2\ (\text{y2Conj})^\dagger\ \text{y2Conj}\ (y_2)^\dagger\Big) \\
&- \frac{2}{3}\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
&- \frac{2}{3}\operatorname{Tr}\Big(\text{y2Conj}\ (\text{y2Conj})^\dagger\ \text{y2Conj}\ (\text{y2Conj})^\dagger\Big) \\
&- \frac{224}{3}\text{lambdaT3GeneralQ000004I1}\,(\text{lambdaT3GeneralQ000004I1})^* \\
&- \frac{4}{3}\text{lambdaT3GeneralQ001102I2}\,(\text{lambdaT3GeneralQ001102I1})^* \\
&- \frac{4}{3}\text{lambdaT3GeneralQ001102I1}(3\,(\text{lambdaT3GeneralQ001102I1})^* + (\text{lambdaT3GeneralQ001102I2})^*) \\
&- \frac{8}{3}\text{lambdaT3GeneralQ110002I1}\,(\text{lambdaT3GeneralQ110002I1})^* \\
&- 2{g_2}^4 \\
&- 24{g_2}^2\lambda_{S_2}^{(A)} \\
&+ 12{\lambda_{12}^{(A)}}^2
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{4{\lambda_{12}^{(\times)}}^2}{3} \\
&+ 3{\lambda_{HS_2}^{(A)}}^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(A)}}{d\ln\mu} = \lambda_{12}^{(A)} &\left(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) + \frac{16\lambda_{12}^{(\times)}}{3}\right) \\
&+ \frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
&- \frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) \\
&- \frac{2}{9}\lambda_5\,(\lambda_5)^* \\
&- \frac{1}{3}\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^* \\
&- 24{g_2}^2\lambda_{12}^{(A)} \\
&- 6{g_1}^2\lambda_{12}^{(A)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)} \\
&+ 3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$\lambda_{12}^{(\times)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(\times)}}{d\ln\mu} = &\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)) \\
&+ \mathrm{lambdaT3GeneralQ001102I2}(2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big) + 4\,(\mathrm{lambdaT3GeneralQ000013I1})^* + 8\,(\mathrm{lambdaT3General} \\
&+ 2\,(\mathrm{lambdaT3GeneralQ001102I2})^*\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big) \\
&- \frac{4}{5}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
&- \frac{4}{5}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) \\
&- \lambda_5\,(\lambda_5)^* \\
&+ 4\mathrm{lambdaT3GeneralQ000013I1}\,(\mathrm{lambdaT3GeneralQ001102I2})^* \\
&+ 8\mathrm{lambdaT3GeneralQ001102I1}\,(\mathrm{lambdaT3GeneralQ001102I2})^* \\
&- \mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^* \\
&+ \frac{51{g_2}^4}{10} \\
&- 24{g_2}^2\lambda_{12}^{(\times)} \\
&- 6{g_1}^2\lambda_{12}^{(\times)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ 10{\lambda_{12}^{(\times)}}^2 \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)} \\
&- 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)} \\
&- 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}
\end{aligned}
$$

$y2Conj$

$$
\begin{aligned}
16\pi^2 \frac{dy2Conj}{d\ln\mu} = {} & 2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big)\,(y_2)_{ij} \\
& + T_\nu^{(2)}\,(\mathrm{y2Conj})_{ij} \\
& + \mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)\,(\mathrm{y2Conj})_{ij} \\
& + \frac{1}{12}(9\left(y_1\ (y_1)^\dagger\ \mathrm{y2Conj}\right)_{ij} + 29\left(y_2\ (y_2)^\dagger\ \mathrm{y2Conj}\right)_{ij} + 29\left(\mathrm{y2Conj}\ (y_2)^\dagger\ y_2\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ \mathrm{y2Conj}\right)_{ij} + 1 \\
& - \frac{9}{2}{g_2}^2\,(\mathrm{y2Conj})_{ij} \\
& - \frac{3}{2}{g_1}^2\,(\mathrm{y2Conj})_{ij}
\end{aligned}
$$

$lambdaT3GeneralQ000004I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ000004I1}{d\ln\mu} = {} & \mathrm{lambdaT3GeneralQ000004I1}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)) \\
& + 2\mathrm{lambdaT3GeneralQ000013I1}\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big) \\
& - \frac{23}{15}\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\ y_2\ (\mathrm{y2Conj})^\dagger\Big) \\
& - 24{g_2}^2\mathrm{lambdaT3GeneralQ000004I1} \\
& + 12\lambda_{S_2}^{(1)}\mathrm{lambdaT3GeneralQ000004I1} \\
& - 8\lambda_{S_2}^{(A)}\mathrm{lambdaT3GeneralQ000004I1} \\
& + 11\mathrm{lambdaT3GeneralQ000013I1}^2 \\
& + 3\mathrm{lambdaT3GeneralQ001102I1}^2 \\
& + 2\mathrm{lambdaT3GeneralQ001102I1lambdaT3GeneralQ001102I2} \\
& + \mathrm{lambdaT3GeneralQ001102I2}^2 \\
& + 2\mathrm{lambdaT3GeneralQ110002I1}^2
\end{aligned}
$$

$lambdaT3GeneralQ000013I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ000013I1}{d\ln\mu} = {} & \lambda_{S_2}^{(1)}(2\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big) + 22\mathrm{lambdaT3GeneralQ000013I1}) \\
& + \mathrm{lambdaT3GeneralQ000004I1}(8\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big) + 88\,(\mathrm{lambdaT3GeneralQ000013I} \\
& + \mathrm{lambdaT3GeneralQ000013I1}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)) \\
& - \frac{46}{15}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (\mathrm{y2Conj})^\dagger\Big) \\
& - \frac{46}{15}\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)
\end{aligned}
$$

$$
\begin{aligned}
&- 24{g_2}^2\text{lambdaT3GeneralQ000013I1} \\
&+ \lambda_{12}^{(1)}(6\text{lambdaT3GeneralQ001102I1} + 2\text{lambdaT3GeneralQ001102I2}) \\
&+ \lambda_{12}^{(\times)}(4\text{lambdaT3GeneralQ001102I1} + 4\text{lambdaT3GeneralQ001102I2}) \\
&+ 4\lambda_{HS_2}^{(1)}\text{lambdaT3GeneralQ110002I1}
\end{aligned}
$$

$lambdaT3GeneralQ001102I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ001102I1}{d\ln\mu} &= \lambda_{12}^{(1)}(2\,\mathrm{Tr}\Big(y_2\ (\text{y2Conj})^\dagger\Big) + 10\text{lambdaT3GeneralQ000013I1} + 8\text{lambdaT3GeneralQ001} \\
&+ \text{lambdaT3GeneralQ001102I1}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\text{y2Conj}\ (\text{y2Conj})^\dagger\Big)) \\
&- \frac{22}{15}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (\text{y2Conj})^\dagger\Big) \\
&+ 8\text{lambdaT3GeneralQ000004I1}(5\,(\text{lambdaT3GeneralQ001102I1})^* + (\text{lambdaT3Gene} \\
&- 24{g_2}^2\text{lambdaT3GeneralQ001102I1} \\
&- 6{g_1}^2\text{lambdaT3GeneralQ001102I1} \\
&+ \lambda_{12}^{(\times)}(4\text{lambdaT3GeneralQ000013I1} + 4\text{lambdaT3GeneralQ001102I2}) \\
&+ 4\lambda_{HS_1}^{(1)}\text{lambdaT3GeneralQ110002I1} \\
&+ \lambda_{S_1}^{(1)}(8\text{lambdaT3GeneralQ001102I1} + 2\text{lambdaT3GeneralQ001102I2}) \\
&+ \lambda_{S_1}^{(A)}(4\text{lambdaT3GeneralQ001102I1} + 2\text{lambdaT3GeneralQ001102I2}) \\
&+ 2\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ001102I1} \\
&+ \lambda_{S_2}^{(A)}(-4\text{lambdaT3GeneralQ001102I1} - 2\text{lambdaT3GeneralQ001102I2}) \\
&+ 2\lambda_5\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$lambdaT3GeneralQ001102I2$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ001102I2}{d\ln\mu} &= \lambda_{12}^{(\times)}(4\,\mathrm{Tr}\Big(y_2\ (\text{y2Conj})^\dagger\Big) + 8\text{lambdaT3GeneralQ000013I1} + 16\text{lambdaT3GeneralQ00 \\
&+ \text{lambdaT3GeneralQ001102I2}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\text{y2Conj}\ (\text{y2Conj})^\dagger\Big)) \\
&- \frac{8}{5}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (\text{y2Conj})^\dagger\Big) \\
&+ 16\text{lambdaT3GeneralQ000004I1}\,(\text{lambdaT3GeneralQ001102I2})^* \\
&- 24{g_2}^2\text{lambdaT3GeneralQ001102I2} \\
&- 6{g_1}^2\text{lambdaT3GeneralQ001102I2} \\
&+ 8\lambda_{12}^{(1)}\text{lambdaT3GeneralQ001102I2} \\
&+ 2\lambda_{S_1}^{(1)}\text{lambdaT3GeneralQ001102I2} \\
&- 2\lambda_{S_1}^{(A)}\text{lambdaT3GeneralQ001102I2} \\
&+ 2\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ001102I2} \\
&+ 2\lambda_{S_2}^{(A)}\text{lambdaT3GeneralQ001102I2} \\
&- 2\lambda_5\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$lambdaT3GeneralQ020101I1$

$$
16\pi^2 \frac{dlambdaT3GeneralQ020101I1}{d\ln\mu} = 2\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_2\ (\text{y2Conj})^\dagger\Big)
$$

$$
\begin{aligned}
&+ \text{lambdaT3GeneralQ020101I1}(T_\nu^{(1)} + T_\nu^{(2)} + \text{Tr}\Big(\text{y2Conj } (\text{y2Conj})^\dagger\Big) + 6\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\\
&+ 4\text{lambdaT3GeneralQ001102I1}\ (\lambda_5)^*\\
&- 2\text{lambdaT3GeneralQ001102I2}\ (\lambda_5)^*\\
&+ 8\text{lambdaT3GeneralQ110002I1}\ (\lambda_5)^*\\
&- \frac{33{g_2}^2\text{lambdaT3GeneralQ020101I1}}{2}\\
&- \frac{9{g_1}^2\text{lambdaT3GeneralQ020101I1}}{2}\\
&+ 2\lambda_{12}^{(1)}\text{lambdaT3GeneralQ020101I1}\\
&- 2\lambda_{12}^{(A)}\text{lambdaT3GeneralQ020101I1}\\
&- 2\lambda_{12}^{(\times)}\text{lambdaT3GeneralQ020101I1}\\
&+ 2\lambda_1\text{lambdaT3GeneralQ020101I1}\\
&+ 4\lambda_{HS_1}^{(1)}\text{lambdaT3GeneralQ020101I1}\\
&- 2\lambda_{HS_1}^{(A)}\text{lambdaT3GeneralQ020101I1}\\
&+ 4\lambda_{HS_2}^{(1)}\text{lambdaT3GeneralQ020101I1}\\
&- 2\lambda_{HS_2}^{(A)}\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$lambdaT3GeneralQ110002I1$

$$
\begin{aligned}
16\pi^2\,\frac{dlambdaT3GeneralQ110002I1}{d\ln\mu} &= \lambda_{HS_2}^{(1)}(2\,\text{Tr}\Big(y_2\ (\text{y2Conj})^\dagger\Big) + 10\text{lambdaT3GeneralQ000013I1} + 8\text{lambdaT3GeneralQ11}\\
&+ \text{lambdaT3GeneralQ110002I1}(2T_\nu^{(2)} + 2\,\text{Tr}\Big(\text{y2Conj } (\text{y2Conj})^\dagger\Big) + 6\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) +\\
&- 2\,\text{Tr}\Big(y_2\ (\text{y2Conj})^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&+ 40\text{lambdaT3GeneralQ000004I1}\ (\text{lambdaT3GeneralQ110002I1})^*\\
&- \frac{33{g_2}^2\text{lambdaT3GeneralQ110002I1}}{2}\\
&- \frac{3{g_1}^2\text{lambdaT3GeneralQ110002I1}}{2}\\
&+ 6\lambda_1\text{lambdaT3GeneralQ110002I1}\\
&+ \lambda_{HS_1}^{(1)}(6\text{lambdaT3GeneralQ001102I1} + 2\text{lambdaT3GeneralQ001102I2})\\
&+ 2\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ110002I1}\\
&- 4\lambda_{S_2}^{(A)}\text{lambdaT3GeneralQ110002I1}\\
&+ 4\lambda_5\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$muT3GeneralS2Sq$

$$
\begin{aligned}
16\pi^2\,\frac{dmuT3GeneralS2Sq}{d\ln\mu} &= m_2^2(2\,\text{Tr}\Big(y_2\ (\text{y2Conj})^\dagger\Big) + 10\text{lambdaT3GeneralQ000013I1})\\
&- 4\,\text{Tr}\Big(M_F\ (\text{y2Conj})^\dagger\ y_2\ (M_F)^\dagger\Big)\\
&+ \text{muT3GeneralS2Sq}(2T_\nu^{(2)} + 2\,\text{Tr}\Big(\text{y2Conj } (\text{y2Conj})^\dagger\Big))\\
&+ 40\text{lambdaT3GeneralQ000004I1}\ (\text{muT3GeneralS2Sq})^*\\
&- 12{g_2}^2\text{muT3GeneralS2Sq}\\
&+ 2\lambda_{S_2}^{(1)}\text{muT3GeneralS2Sq}\\
&- 4\lambda_{S_2}^{(A)}\text{muT3GeneralS2Sq}
\end{aligned}
$$

$$+ m_1^2(6\text{lambdaT3GeneralQ001102I1} + 2\text{lambdaT3GeneralQ001102I2})$$